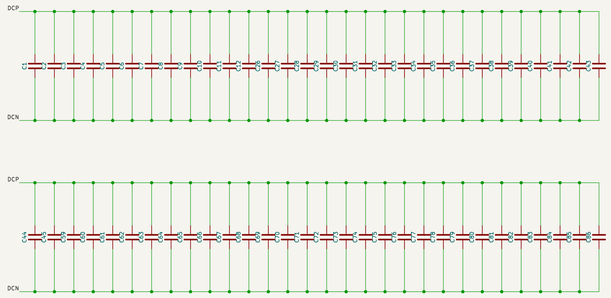
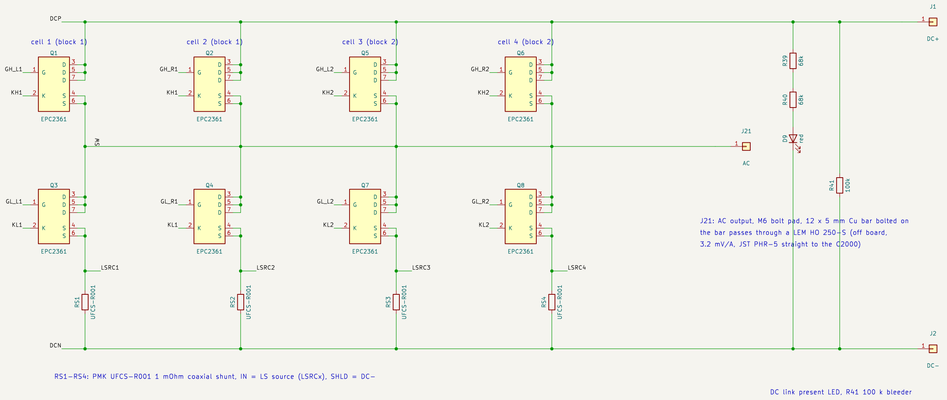
POWER STAGE AND DC LINK (block 1: Q1–Q4, block 2: Q5–Q8; one cell = HS + LS transistor + UFCS shunt in the LS source; AC of all four cells joined on the PCB)



DC link, 60 capacitors: 40 x 1uF 100V X7S, 20 x 22uF 100V X7S 2220 (1 uF 0805: bypass at the HS drains of each cell, top + bottom; 22 uF 2220: bank)



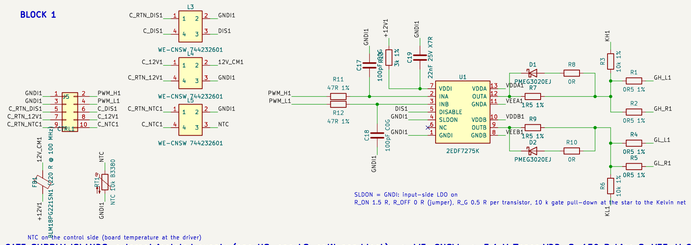
RS1–RS4: PMK UFCS-R001 1 mOhm coaxial shunt, IN = LS source (LSRCx), SHLD = DC−

J21: AC output, M6 bolt pad, 12 x 5 mm Cu bar bolted on the bar passes through a LEM HO 250-S (off board, 3.2 mV/A, JST PHR-5 straight to the C2000)

DC link present LED, R41 100 k bleeder

GATE DRIVER BLOCKS: control header (chokes on DIS / 12 V / NTC), 2EDF7275K input filters, R_ON / D + R_OFF, star, R_G per transistor (block 1 drives Q1–Q4, block 2 drives Q5–Q8)

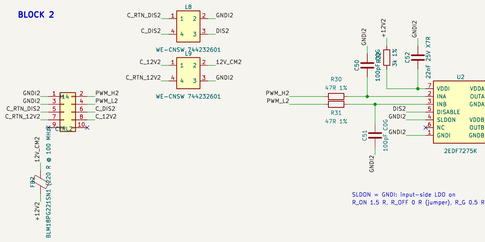
BLOCK 1

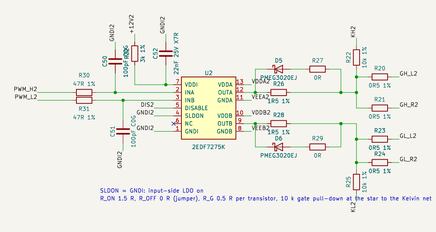


NTC on the control side (board temperature at the driver)

SLDON = GNDI: input-side LDO on
R_ON 1.5 R, R_OFF 0 R (jumper), R_G 0.5 R per transistor, 10 k gate pull-down at the star to the Kelvin net

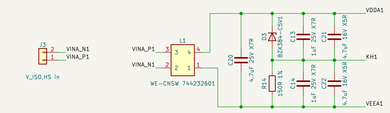
BLOCK 2



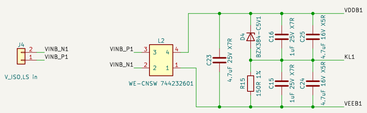


SLDON = GNDI: input-side LDO on
R_ON 1.5 R, R_OFF 0 R (jumper), R_G 0.5 R per transistor, 10 k gate pull-down at the star to the Kelvin net

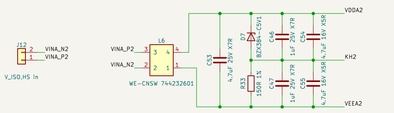
GATE SUPPLY ISLANDS: external isolated supply (one HS, one LS, split per block) –> WE-CNSW –> 5.1 V Zener VDD–S, 150 R bias S–VEE: V_GS,on = +5.1 V, V_GS,off = −(V_ISO − 5.1 V)



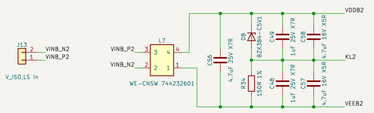
J3 / L1: HS block 1



J4 / L2: LS block 1



J12 / L6: HS block 2



J13 / L7: LS block 2

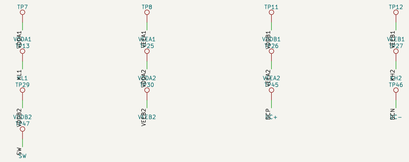
MEASUREMENT PORTS (bottom side, at each cell): U.FL V_GS HS / LS (0 R from the gate, shield = Kelvin); MMCX V_DS LS (Wurth WR-MMCX, 170 V rms; centre = SW at the LS drain, shield = LS source LSRCx, device side of the shunt; for a rated isolated probe, e.g. IsoVu TIVP + TIVPMX50X)



MOUNTING: H1–H4 board corners (insulated); no heatsink (double-pulse board)



TEST POINTS: loop test points Adafruit 3825 / Keystone 5000 (red), 1.02 mm hole, inserted in the lab; SMD test pads and V_GS / V_DS spring pads: see r



Nets: DCP = DC+, DCN = DC−, SW = joined switch node of all four cells. KH1/KH2 and KL1/KL2 are the Kelvin source nets (pad 2 of each EPC2361): they meet the power source only inside the package.

LSRC1–LSRC4: LS power source of each cell = UFCS IN; the shunt SHLD side is DC−. The LS gate drive and the LS supply are referenced to the Kelvin net KLx, not to DC−.

Block suffix 1/2 on every driver-side net (GNDI1, VDDA2, …). Control side (CTRLx 2 x 5 header): PWM_H and PWM_L straight to the 47 R / 100 pF input filters (returns to GNDI, no choke: no added delay); DIS, 12 V and (block 1) NTC each with their own return C_RTN_* through one WE-CNSW common-mode choke per pair; 12 V then through a ferrite bead. The control side is a top-layer-only area with a 1200 V barrier (4 mm outer, 3 mm L2) to all power copper; see reports/pcb-rules/PCB_rules_and_checklist.md.

No short-circuit protection on this board. Closed-loop control from a TI TMS320F28379D over the CTRL headers.

generated by scripts/make_kicad_sch_revb.py; netlist identical to the PCB (check_sch_vs_pcb.py)
4 x EPC2361 per switch, 2 x 2EDF7275K, UFCS shunt per cell, no heatsink, external isolated supplies

KTH

Sheet: /	
File: spb-bb-revb-dpt.kicad_sch	
Title: SPB leg, Rev B, double-pulse board	
Size: A1	Date: 2026-10-14
KiCad E.D.A. kicad-cli 7.0.10	**Rev: B**
	Id: 1/1